

Software Requirements Specification

For

Kheti Saathi: An Intelligent Farming Assistance System for Crop Recommendation and Plant Disease Detection Using Machine Learning Techniques

14/03/2026

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1. Introduction

1.1 Purpose

The purpose of this Software Requirement Specification (SRS) document is to provide a detailed description of the requirements, functionalities, and design considerations of the Kheti Saathi system. Kheti Saathi is an artificial intelligence-based mobile application designed to assist farmers in making better agricultural decisions using modern technology. The system integrates machine learning and deep learning techniques to provide intelligent recommendations related to crop selection and plant disease detection.

This document describes the system's objectives, overall structure, functional requirements, non-functional requirements, and technologies used to develop the application. The main goal of this document is to clearly define how the system will operate and what functionalities it will provide to its users. It also serves as a guideline for developers, project evaluators, and stakeholders to understand the working and capabilities of the proposed system.

1.2 Scope

Kheti Saathi is designed as a smart farming assistant that helps farmers improve agricultural productivity and reduce crop losses. The application provides intelligent recommendations based on soil nutrients, environmental conditions, and plant health analysis. The system aims to bridge the gap between traditional farming practices and modern technological solutions.

The application provides two major functionalities: crop recommendation and plant disease detection. The crop recommendation module analyzes soil nutrient values and environmental conditions such as nitrogen, phosphorus, potassium, temperature, humidity, pH level, and rainfall to suggest the most suitable crop that can be cultivated in those conditions. The plant disease detection module allows farmers to identify diseases affecting their crops by analyzing images of plant leaves using deep learning techniques.

The system is implemented as a mobile application using the Flutter framework, allowing farmers to interact with the system through a simple and user-friendly interface. The goal of the application is to provide farmers with reliable, accessible, and easy-to-use technology that supports better decision-making in agriculture.

1.3 Overview

This document describes the structure and requirements of the Kheti Saathi system. It includes sections that explain the purpose of the system, system description, functional requirements, non-functional requirements, system architecture, and technologies used for development. The document provides a clear understanding of how the system will operate and how the different modules of the system interact with each other.

2. Overall Description

2.1 Product Perspective

Kheti Saathi is a standalone mobile application designed to assist farmers in agricultural decision-making. The system integrates machine learning and deep learning models to provide crop recommendations and plant disease detection functionalities. The application serves as a decision support tool that helps farmers choose suitable crops and identify plant diseases at an early stage.

The system consists of multiple components including the user interface, machine learning models, and processing modules. The mobile application interface allows farmers to input soil parameters and upload leaf images, while the backend processing module analyzes the data using trained models to generate predictions.

2.2 Product Functions

The Kheti Saathi system provides several important functions that support farmers in managing agricultural activities. The crop recommendation module analyzes soil nutrients and environmental conditions to predict the most suitable crop for cultivation. The plant disease detection module analyzes leaf images to identify plant diseases using deep learning models.

The system processes user input data and provides prediction results that help farmers make informed decisions regarding crop selection and plant health management. These functions collectively improve agricultural productivity and reduce crop losses caused by improper crop selection and plant diseases.

2.3 User Characteristics

The primary users of the Kheti Saathi system are farmers and agricultural workers who may not have extensive technical knowledge. Therefore, the system is designed with a simple and user-friendly interface so that users can easily interact with the application.

Farmers using the system are expected to have basic knowledge of smartphones and mobile applications. The interface is designed in such a way that users can easily enter soil parameters and upload leaf images without difficulty.

2.4 Constraints

The Kheti Saathi system may have certain limitations related to data accuracy and device capabilities. The accuracy of crop recommendation depends on the quality of the dataset used to train the machine learning model. Similarly, the accuracy of plant disease detection depends on the quality of the leaf image provided by the user.

Another constraint is the processing capability of the mobile device on which the application is running. Devices with limited processing power may experience slower prediction times when running machine learning models.

3. System Features

3.1 Crop Recommendation Module

The crop recommendation module is designed to help farmers select the most suitable crop for cultivation based on soil and environmental conditions. The user provides input values such as nitrogen level, phosphorus level, potassium level, temperature, humidity, soil pH, and rainfall. The system processes these parameters using a trained machine learning model to determine the most appropriate crop that can be grown under the given conditions.

The machine learning algorithm used in this module analyzes patterns in agricultural datasets and predicts crops that are most likely to produce high yields under specific soil conditions. This feature helps farmers make informed decisions about crop selection and improves overall farming efficiency.

3.2 Plant Disease Detection Module

The plant disease detection module helps farmers identify diseases affecting their crops by analyzing images of plant leaves. The user uploads or captures a leaf image through the mobile application. The system processes the image using image preprocessing techniques such as resizing and normalization.

The processed image is then analyzed by a convolutional neural network (CNN) model trained on plant leaf image datasets. Based on the features extracted from the image, the model predicts whether the plant is healthy or affected by a particular disease. The result is displayed to the user along with the predicted disease name.

4. External Interface Requirements

4.1 User Interface

The user interface of the Kheti Saathi system is developed using the Flutter framework. The interface is designed to be simple, intuitive, and easy to navigate. The application includes input forms for soil parameters and options to upload plant leaf images for disease detection.

Prediction results are displayed clearly so that farmers can easily understand the recommendations provided by the system.

4.2 Hardware Interface

The Kheti Saathi application runs on Android smartphones that support camera functionality and basic processing capabilities. The device camera is used for capturing plant leaf images for disease detection.

4.3 Software Interface

The system integrates several software technologies including Flutter for mobile application development and Python-based machine learning libraries for model development. Libraries such as TensorFlow, Scikit-learn, NumPy, Pandas, and OpenCV are used for implementing the machine learning and deep learning components of the system.

5. Functional Requirements

The Kheti Saathi system must allow users to enter soil parameters including nitrogen, phosphorus, potassium, temperature, humidity, pH level, and rainfall. The system should process these values using the trained machine learning model and generate crop recommendations based on the analysis.

The system must also allow users to capture or upload images of plant leaves. Once the image is provided, the system should preprocess the image and analyze it using the deep learning model to detect plant diseases. The system should then display the detected disease or indicate if the plant is healthy.

Additionally, the system should present prediction results clearly so that farmers can easily understand the recommendations provided by the application.

6. Non-Functional Requirements

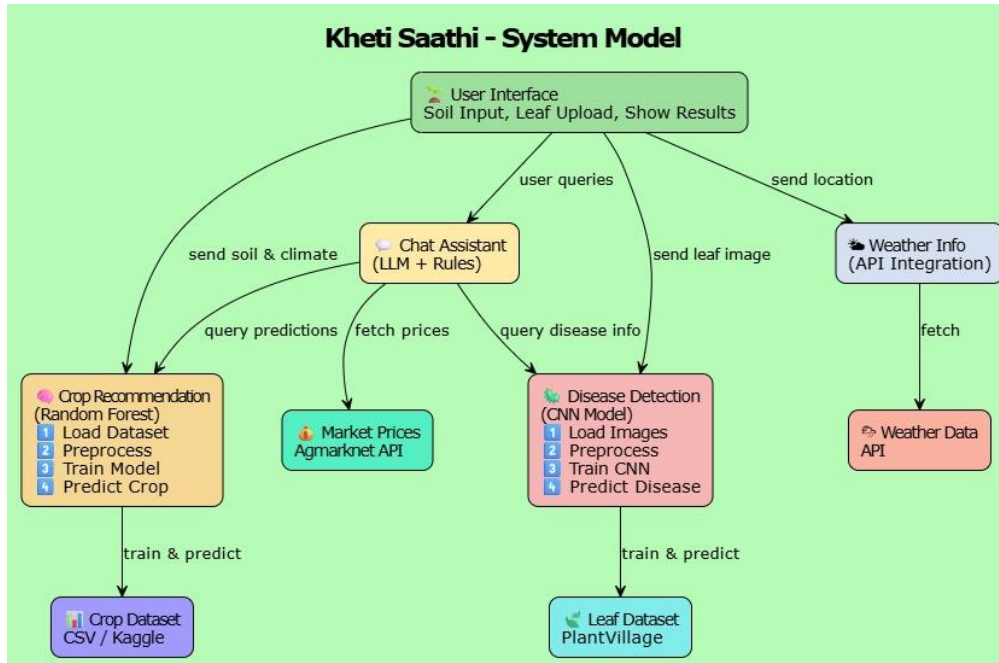
The Kheti Saathi application must provide fast and efficient predictions so that farmers can receive results without significant delay. The system should maintain reliability by producing accurate predictions based on well-trained machine learning and deep learning models.

The application must also be designed with usability in mind, ensuring that the interface is simple and easy to understand for users who may not have extensive technical knowledge. The system should maintain user privacy and should not store personal data or images without the user's permission.

7. System Architecture

The architecture of the Kheti Saathi system consists of three main layers: the user interface layer, the processing layer, and the machine learning models. The user interacts with the system through the mobile application interface. The application collects input data such as soil parameters or plant leaf images and sends this information to the processing module.

The processing module analyzes the input data using trained machine learning and deep learning models. The crop recommendation model predicts the most suitable crop based on soil and environmental parameters, while the plant disease detection model analyzes leaf images to identify diseases. The final prediction results are then displayed to the user through the application interface.



8. Technologies Used

The Kheti Saathi system is developed using a combination of modern mobile development and machine learning technologies. The mobile application interface is built using the Flutter framework, which enables the development of cross-platform mobile applications using the Dart programming language.

The machine learning components of the system are implemented using Python libraries. The crop recommendation system uses machine learning algorithms implemented with Scikit-learn, while the plant disease detection module uses deep learning models developed using TensorFlow and Keras. Additional libraries such as NumPy, Pandas, OpenCV, and Matplotlib are used for data processing, image processing, and visualization tasks.

9. Methodology

The methodology of the Kheti Saathi system focuses on the implementation of machine learning and deep learning techniques to perform crop recommendation and plant disease detection. The system consists of two primary analytical components: the crop recommendation model based on the Random Forest algorithm and the plant disease detection model based on Convolutional Neural Networks (CNN). These models are trained using agricultural datasets and are integrated into the mobile application to provide predictions based on user inputs.

9.1 Data Collection

The first step in the methodology is data collection. Agricultural datasets containing information about soil nutrients, environmental conditions, and crop types are used to train the crop recommendation model. These datasets include parameters such as nitrogen, phosphorus,

potassium, temperature, humidity, soil pH, rainfall, and the corresponding crop that grows best under those conditions.

For plant disease detection, image datasets containing leaf images of different crops and diseases are used. These datasets include images of both healthy and diseased plant leaves, enabling the deep learning model to learn patterns associated with various plant diseases.

	A	B	C	D	E	F	G	H
1	N	P	K	temperatu	humidity	ph	rainfall	label
2	90	42	43	20.87974	82.00274	6.502985	202.9355	rice
3	85	58	41	21.77046	80.31964	7.038096	226.6555	rice
4	60	55	44	23.00446	82.32076	7.840207	263.9642	rice

<https://www.kaggle.com/code/niteshhalai/crop-recommendation-dataset>

9.2 Data Preprocessing

Before training the models, the collected datasets undergo preprocessing to ensure data quality and consistency. In the crop recommendation dataset, missing values and inconsistent data entries are handled, and the dataset is normalized to improve model performance.

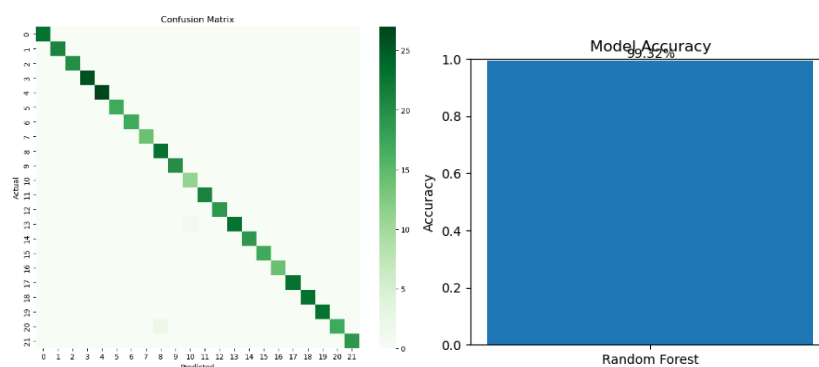
For the plant disease detection dataset, image preprocessing techniques are applied. These include resizing images to a fixed dimension, normalization of pixel values, and augmentation techniques such as rotation and flipping to increase the diversity of the training data.

9.3 Crop Recommendation Using Random Forest

The crop recommendation module uses the Random Forest machine learning algorithm to analyze soil parameters and predict suitable crops. Random Forest is an ensemble learning algorithm that builds multiple decision trees during training and combines their outputs to produce accurate predictions.

The model is trained using the agricultural dataset containing soil nutrient values and environmental conditions. During the prediction stage, the user inputs parameters such as nitrogen, phosphorus, potassium, temperature, humidity, pH level, and rainfall into the application. The trained Random Forest model analyzes these inputs and predicts the most suitable crop for cultivation.

Random Forest is selected for this task because of its high accuracy, robustness to overfitting, and ability to handle complex relationships between multiple input parameters.



9.4 Plant Disease Detection Using Convolutional Neural Networks (CNN)

The plant disease detection module is implemented using a Convolutional Neural Network (CNN), which is a deep learning architecture specifically designed for image classification tasks. CNN models are capable of automatically extracting features from images and identifying patterns associated with different classes.

In this system, the CNN model is trained using plant leaf image datasets containing images of healthy leaves and leaves affected by various diseases. The model learns to recognize visual patterns such as spots, discoloration, and texture changes that indicate specific plant diseases.

When a user uploads a leaf image through the application, the image is processed and passed through the trained CNN model. The model analyzes the image and predicts whether the plant is healthy or affected by a particular disease. The prediction result is then displayed to the user.

9.5 System Integration

The trained machine learning and deep learning models are integrated into the Kheti Saathi mobile application. The Flutter-based user interface allows users to input soil parameters and upload leaf images. The application processes the input data and passes it to the corresponding models for analysis.

The crop recommendation model generates predictions based on soil parameters, while the disease detection model processes leaf images to identify plant diseases. The results from these models are then displayed to the user through the mobile application interface.

10. Future Enhancements

The Kheti Saathi system can be further improved by integrating additional features that enhance agricultural decision-making. Future improvements may include integration with real-time weather forecasting services, fertilizer recommendation systems, crop market price information, and voice-based interaction to assist farmers who may have difficulty using text-based interfaces. These enhancements would make the system more comprehensive and beneficial for farmers.

11. Conclusion

Kheti Saathi is an intelligent agricultural assistance system developed to support farmers by utilizing modern artificial intelligence technologies. The system integrates machine learning algorithms for crop recommendation and deep learning models for plant disease detection, providing farmers with valuable insights that assist in making informed agricultural decisions.

By analyzing soil parameters and environmental conditions, the crop recommendation module helps farmers identify the most suitable crops for cultivation. At the same time, the plant disease detection module enables farmers to detect plant diseases early by analyzing leaf images. Early detection allows farmers to take timely preventive measures and reduce potential crop losses.

The mobile application interface ensures that the system is accessible and easy to use, even for individuals with limited technical knowledge. By combining advanced AI technologies with a user-friendly interface, the system bridges the gap between modern technological advancements and traditional farming practices.

Overall, Kheti Saathi demonstrates how artificial intelligence can be applied in the agricultural sector to improve productivity, reduce crop losses, and promote sustainable farming practices. With future enhancements and additional features, the system has the potential to become a powerful digital assistant that significantly benefits farmers and contributes to the advancement of smart agriculture.

References:

Below are recent research papers and sources related to plant disease detection and crop recommendation using machine learning, which support the concept of the Kheti Saathi project.

1. Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016).
Using Deep Learning for Image-Based Plant Disease Detection. *Frontiers in Plant Science*.
This research demonstrated that deep learning models trained on large datasets of plant leaf images can accurately identify plant diseases across multiple crop species.
2. Kim, T. et al. (2024).
ANFIS Fuzzy Convolutional Neural Network Model for Leaf Disease Detection. *Frontiers in Plant Science*.
This research proposes an advanced CNN-based approach that improves plant leaf disease detection accuracy by analyzing visual patterns in plant leaves.
3. Shafay, M. et al. (2025).
Recent Advances in Plant Disease Detection Using Artificial Intelligence.
The study highlights how AI-based disease detection systems can improve agricultural productivity and support sustainable farming practices.
4. Fu, Y. et al. (2024).
A Lightweight CNN Model for Pepper Leaf Disease Recognition.
This paper introduces a CNN model designed to detect plant diseases efficiently and demonstrates the effectiveness of deep learning in crop monitoring.
5. Prity, F. S. et al. (2024).
A Machine Learning Approach to Crop Recommendation Systems.
The research proposes a crop recommendation system that uses soil properties, climatic conditions, and historical agricultural data to suggest suitable crops.
6. Shastri, S. et al. (2025). **Advancing Crop Recommendation Systems with Supervised Machine Learning.**

The research demonstrates how machine learning algorithms can analyze agricultural data to recommend suitable crops and improve yield prediction.

7. Kanakala, S. & Ningappa, S. (2025).

Detection and Classification of Diseases in Multi-Crop Leaves using CNN and LSTM Models.

This study demonstrates that deep learning models can detect multiple plant diseases with high accuracy using large datasets of leaf images.